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The PCL-R Assessment of Psychopathy

Robert D. Hare

Introduction

Much has happened in the field of psychopathy since I wrote a chapter (Hare, 2007) for the previous edition of the *International Handbook of Psychopathic Disorders and the Law* (Felthous & Saß, 2007). Although based partly on the previous chapter, the current chapter draws heavily on the extensive research literature that has emerged over the past decade, and on several recent and extensive discussions of psychopathy (Hare, 2016; Hare, Black, & Walsh, 2013; Hare, Neumann, & Mokros, 2018; Mokros et al., 2015).

Psychopathy is a clinical construct that includes a cluster of interpersonal, affective, lifestyle, and antisocial traits and behaviors, including deception, manipulation, irresponsibility, impulsivity, stimulation-seeking, poor behavioral controls, shallow affect, a lack of empathy, guilt, or remorse, and a range of unethical and antisocial behaviors, not necessarily criminal. Among the most devastating features of psychopathy are a callous disregard for the rights of others and high risk for a variety of predatory and aggressive behaviors (Hare, 1999).

The focus in this chapter is on the most widely accepted clinical/forensic measure of psychopathy, the Psychopathy Checklist-Revised (PCL-R; Hare, 1991, 2003). I also briefly describe its direct derivatives, the Psychopathy Checklist: Screening Version (PCL: SV; Hart, Cox, & Hare, 1995), the Psychopathy Checklist: Youth Version (PCL: YV; Forth, Kosson, & Hare, 2003), and the Antisocial Process Screening Device (APSD; Frick & Hare, 2001). Extensive evidence supports the reliability and validity of each of these instruments. Psychopathy is related to, but not synonymous with dissocial or antisocial personality disorder (ASPD) as described, respectively, in the International Classification of Diseases (ICD-10; World Health Organization, 2004) and the fifth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5; American Psychiatric Association, 2013). I provide more details in the section on ICD-10 and DSM-5, below. Extensive discussions are available elsewhere (e.g., Crego & Widiger, 2015; Hare, Neumann, & Widiger, 2012; Saß & Felthous, 2007; Chapters 1–3 and 28 this volume).

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Other chapters in the two volumes of this international handbook provide detailed accounts of the research, applications, issues, and debates concerning the clinical and forensic use of the psychopathy construct.

Psychopathy

Throughout recorded history, there have been many descriptions of individuals whose traits and behaviors were remarkably consistent with current clinical conceptions of psychopathy. However, it was only in the last several hundred years that psychopathy began to take shape as a formal psychiatric disorder, reflected in the clinical investigations and speculations of European and North American psychiatrists and psychologists (see Berrios, 1996; Cleckley, 1976; Gacono, 2016; Hare, 2003; Hervé, 2007; Meloy, 1988; Millon, Simonsen, Birket-Smith, & Davis, 1998; Saß & Felthous, 2007). The writings of Cleckley (e.g., 1941, 1976) have been particularly influential in providing a framework for the research conducted over the last 50 years (Crego & Widiger, 2016; Hare & Neumann, 2008; Lilienfeld, Watts, Smith, Berg, & Latzman, 2015; Lilienfeld, Watts, Smith, Patrick, & Hare, 2018). In the last quarter century, there has been a dramatic increase in basic and applied research on psychopathy, due in large part to the development and validation of new assessment tools, particularly the PCL-R and its derivatives.

The basic research on psychopathy includes empirical investigations of etiology, measurement, developmental course, cross-cultural generalizability, issues related to behavioral genetics, and cognitive/affective neuroscience. Much of the applied work involves matters of interest to forensic psychology and psychiatry, including assessments of risk for violence and recidivism, parole decisions, civil commitment and dangerous offender proceedings, and evaluations of treatment suitability. These and related topics are covered in detail throughout the two volumes of this international handbook.

Important forensic developments include the extension of the concept of psychopathy to the community (e.g., Miller & Lynam, 2015b; Neumann & Hare, 2008; Neumann, Hare, & Pardini, 2015), the workplace (Babiak & Hare, 2019; Babiak, Neumann, & Hare, 2010; Mathieu, Babiak, & Hare, Chapter 27 this volume), and law enforcement (O'Toole, 2007; Perri, 2016). With respect to the latter, the FBI *Law Enforcement Bulletin* devoted an entire issue to psychopathy in 2012 (https://leb.fbi.gov/2012/july/leb-july-2012). Because of its pervasive influence, I once described psychopathy as the single most important clinical construct in the criminal justice system (Hare, 1996). Monahan (2006) referred to it as the most important forensic concept of the early twenty-first century. Delisi (2009, 2018) argued that psychopathy is so intertwined with criminal behavior that it provides a unified theory of crime. His approach, as is mine, is interdisciplinary in nature.

Is Psychopathy a Disorder?

Although traditionally considered to be a personality disorder (Millon, Simonson, & Birket-Smith, 1998, p. 28), some investigators view psychopathy as an evolutionary adaptive life strategy rather than as a disorder (e.g., Book, & Quinsey, 2003; Brazil, & Forth, 2019; Glenn,

Kurzban, & Raine, 2011; Glenn & Raine, 2014; Lalumière, Mishra, & Harris, 2008; Mealey, 1995). More recently, personality theorists have conceptualized psychopathy as an extreme variant of general personality dimensions (Hare et al., 2012; Miller, & Lynam, 2015c). Other investigators see psychopathy as involving a variety of neurological and cognitive anomalies, usually in the form of "deficits." This latter view has important implications for criminal culpability, and is covered in detail elsewhere (Hamilton, Hiatt Racer, & Newman, 2015; Kiehl, & Sinnott-Armstrong, 2013; Malatesti & McMillan, 2010; Saks, Schweitzer, Aharoni, & Kiehl, 2014; Umbach, Berryessa, & Raine, 2015). I believe that caution is required when interpreting functional and structural differences between psychopathic and other individuals as indicative of a deficit or malfunction. At the very least

> ... we should consider the possibility that the actions of psychopaths reflect cognitive, affective, and behavioural processes and strategies that are different from those of other people, but for reasons other than neuropathology or deficit, in the traditional medical and psychiatric sense of the terms. (Hare, 2013, pp. vii–ix)

Etiology, Dynamics, and Conceptual Boundaries

There is a rather consistent clinical and empirical tradition concerning the affective, interpersonal, and behavioral attributes of psychopathy. The etiology, dynamics, and conceptual boundaries of the construct remain the subjects of debate and research, but are beyond the scope of this chapter. However, a few comments are in order.

Psychopathy is not solely, or even primarily, a product of social and environmental forces and influences. It is clear that genetic factors play an important role in the formation of the personality traits and temperament considered essential to the disorder (Viding & McCrory, 2012; Waldman, Rhee, LoParo. & Park, 2018). However, its lifelong expression is a product of complex interactions between biological/temperamental predispositions and social forces (MacDonald & Iacono, 2006). Certainly, the traits and behaviors that help to define adult psychopathy begin to manifest themselves early in childhood (Frick, 2009; Lynam & Gudonis, 2005; Lynam & Vachon, 2012; Viding & McCrory, 2012).

There are many extensive discussions of current models of psychopathy based on cognitive/affective neuroscience (Blair, 2005, 2013; Ermer, Cope, Nyalakanti, Calhoun, & Kiehl, 2012; Hamilton et al., 2015; Lushing, Gaudet, & Kiehl, 2016; Seara-Cardoso & Viding, 2015). As described by Hamilton et al. (2015), emotion-based (affective) models and attentional (cognitive) models dominate cognitive/affective research on psychopathy. They provide reviews of both types of models and argue that the apparent dichotomy between them is specious. The authors present an *impaired information* (II) model that integrates the research findings concerning the affective and cognitive aspects of psychopathy:

> We propose that at the core of psychopathy lies a fundamental deficit in perceptual integration. Specifically, our II framework states that failure to rapidly bind components of multidimensional stimuli in psychopathy creates a perceptual bottleneck resulting in unelaborated mental representations and the development of abnormal topography in associative neural networks. (Hamilton et al., 2015, p. 777)

They note (p. 770) that a central premise of II theory is that psychopathic individuals are "wired up" differently, quoting Hare, Williamson, and Harpur (1988, p. 87). In effect, there is impaired integration of affective and cognitive mechanisms and circuits. Several decades ago, I proposed a much less elaborate version of this model, based on the available literature and the work of Damasio (1994). I raised the possibility that "psychopathy is associated with anomalies in cortical/subcortical structures and functional circuits responsible for the integration of cognition, affect, and behavior" (Hare, 1998a, p. 117).

Whatever its etiology or psychobiological nature, psychopathy clearly presents society with a serious problem (Delisi, 2018; Hare, 1998b; Reidy et al., 2015). Although many (perhaps most) psychopaths do not come into formal contact with the criminal justice system, their features clearly place them at high risk for violating the rules of society. As Silver, Mulvey, & Monahan (1999) put it, "Psychopathy's defining characteristics, such as impulsivity, criminal versatility, callousness, and lack of empathy or remorse make the conceptual link between violence and psychopathy straightforward" (p. 244). The problem is to assess psychopathy as professionally and accurately as possible, particularly in situations where such an assessment has serious implications for both individuals and society.

Psychopathy Checklist-Revised (PCL-R)

The purpose of the PCL-R was to measure the construct of psychopathy. It describes psychopathy as a constellation of interpersonal, affective, lifestyle, and antisocial traits and behaviors (see Table 4.1, left panel). The extensive evidence for the reliability and validity of the first edition (1991) of the PCL-R led the Buros 12th Mental Measurements Yearbook to describe it as "state of the art ... both clinically and in research use" (Fulero, 1995, pp. 453–454). Following publication of the second edition of the PCL-R (Hare, 2003), the 16th Mental Measurements Yearbook referred to it as "the gold standard for the assessment of psychopathy" (Acheson, 2005, pp. 429–431). As noted by Westen and Weinberger (2004, p. 599):

> Virtually all current research on psychopathy presupposes the observations of a brilliant clinical observer [Cleckley, 1941] whose clinical immersion among psychopaths over 60 years ago still provides the foundation for the measure [the PCL-R] considered the gold standard in psychopathy research.

The PCL-R is based firmly on a widely accepted clinical and empirical tradition and serves as a nexus or anchor for recent research and discussions concerning the nomological network of psychopathy (Benning, Patrick, Salekin, & Leistico, 2005; Crego & Widiger, 2015; Hare et al., 2012; Vachon, Lynam, Loeber, & Stouthamer-Loeber, 2012). In many jurisdictions, the PCL-R is part of the "best practices" protocols (e.g., Khiroya, Weaver, & Maden, 2009). It is clear that this instrument has helped to fill a diagnostic and assessment void by providing researchers and clinicians with a common metric that has demonstrated reliability and validity in an array of populations and contexts. The past 20 years have seen a sharp rise in its use in the criminal justice system worldwide (DeMatteo & Edens, 2006; DeMatteo et al., 2014; Guy, Kusaj, Packer, & Douglas, 2015; Hurducas, Singh, de Ruiter, & Petrila, 2014; Neal & Grisso, 2014; Singh, Bjørkly, & Fazel, 2016; Singh et al., 2014; Walsh & Walsh, 2006).

Table 4.1 Items, Factors, and Facets in the Hare PCL Scales

PCL-R	PCL: YV	PCL: SV
Factor 1 (F1)		*Part 1 (Factor 1)*
Interpersonal	Interpersonal	Interpersonal
1. Glibness/superficial charm	1. Impression management	1. Superficial
2. Grandiose sense of self worth	2. Glibness/superficial charm	2. Grandiose
4. Pathological lying	4. Pathological lying	3. Deceitful
5. Conning/manipulative	5. Manipulation for personal gain	
Affective	Affective	Affective
6. Lack of remorse of guilt	6. Lack of remorse	4. Lacks remorse
7. Shallow affect	7. Shallow affect	5. Lacks empathy
8. Callous/Lack of empathy	8. Callous/Lack of empathy	6. Does not accept responsibility
16. Failure to accept responsibility	16. Failure to accept responsibility	
Factor 2 (F2)		*Part 2 (Factor 2)*
Lifestyle	Behavioral	Lifestyle
3. Need for stimulation	3. Stimulation-seeking	7. Impulsive
9. Parasitic lifestyle	9. Parasitic orientation	9. Lacks goals
13. No realistic, long-term goals	13. Lack of goals	10. Irresponsibility
14. Impulsivity	14. Impulsivity	
15. Irresponsibility	15. Irresponsibility	
Antisocial	Antisocial	Antisocial
10. Poor behavioral controls	10. Poor anger control	8. Poor behavioral controls
12. Early behavioral problems	12. Early behavior problems	11. Adolescent antisocial behavior
18. Juvenile delinquency	18. Serious criminal behavior	12. Adult antisocial behavior
19. Revoke conditional release	19. Serious violations of release	
20. Criminal versatility	20. Criminal versatility	

PCL-R = Psychopathy Checklist-Revised; PCL: SV = Psychopathy Checklist: Screening Version; PCL: YV = Psychopathy Checklist: Youth Version. The items are reprinted with permission of the copyright holder, Multi-Health Systems. Note that the item titles cannot be scored without reference to the formal criteria contained in the published manuals. PCL-R items 11 (Promiscuous sexual behavior) and 17 (Many short-term marital relationships) contribute to the Total score but do not load on any factors. PCL: YV items 11, Impersonal sexual behavior, and 17, Unstable interpersonal relationships, contribute to the Total score but do not load on any factor. F1 and F2 are the original PCL-R factors, but with the addition of item 20.

Although the PCL-R is a measure of a clinical construct, not a risk tool, several international surveys indicate that it is one of the most widely used psychological instruments in forensic psychiatry and psychology (Hurducas et al., 2014; Neal & Grisso, 2014; Singh et al., 2016). As described by Storey, Hart, Cooke, & Michie (2016, pp. 144–145):

> The PCL-R is among the most commonly used tests in the practice of forensic mental health internationally (Neal & Grisso, 2014; Singh et al., 2014). This situation is unlikely to change in the near future, given continued efforts to translate and validate the test and the absence of an emerging competitor. But responsible use of the PCL-R requires us to appreciate its potential limitations as indicated by past research ...

Development

Detailed descriptions of the origins and development of the PCL-R are available elsewhere (Hare, 1991, 2003; Hare et al., 2013; Hare & Neumann, 2008). Briefly, in the 1960s and 1970s, researchers used a variety of assessment and diagnostic procedures, most of which were conceptually and empirically unrelated to one another (Hare, 1985). Because there was no reliable, valid, and generally acceptable method for the assessment of psychopathy, it was difficult or impossible to compare results from different researchers and studies. This prompted my research staff, students, and me to attempt development of a common metric for the assessment of psychopathy by combining personality traits and antisocial behaviors, in line with clinical tradition. These efforts resulted in a 22-item scale (Hare, 1980, 1985), later referred to as the PCL. Factor analyses of the PCL items, each scored on a 3-point scale, led to a solution with two correlated factors, labeled as Factor 1: *Selfish, callous, and remorseless use of others*; and Factor 2: *Chronically unstable and antisocial lifestyle, or Social deviance* (Harpur, Hakstian, & Hare, 1988; Harpur, Hare, & Hakstian, 1989). In 1980, we began to disseminate a mimeographed manual for use by other investigators (Hare & Frazelle, 1980). The introduction of the PCL and its two-factor structure led to a sharp increase in research on psychopathy, its dimensions, and their correlates. As put by Lilienfeld (2018, p. 80):

> Were I forced to nominate the most important psychopathy article over the past three decades, my vote would go to Harpur et al. (1989). This seminal publication was the first to highlight the dual points that (a) psychopathy comprises more than one dimension, and (b) these subfactors often display psychologically important differential correlates.

It is important to note that although Cleckley was a major inspiration for the content of the PCL, he was not alone in this regard. As I noted elsewhere (Hare et al., 2018, p. 48; also see Crego & Widiger, 2015; Hare & Neumann, 2008):

> Hare did not base the items on a simple, uncritical acceptance and application of the 16 characteristics listed in Cleckley's (1976) Clinical Profile of psychopathy. Rather, they were based on a deep appreciation of the rich clinical material he provided (some an interpretation and integration of other clinical writings), 15 years of experience and

empirical research by Hare and his colleagues and students, and theoretical and empirical studies by other clinicians and researchers, all before the PCL was conceived and developed.

Editions

Comments and concerns from fellow researchers indicated that it was important to make a number of improvements to the PCL. In 1985, I began to circulate a draft version of the revision throughout the research community. Subsequently, the draft was fine-tuned and the scoring criteria clarified in order to make the manual (and the instrument) easier for other investigators to use. A formal manual appeared as the Hare PCL-R (Hare, 1991). Although some commentators were concerned that formal measures of low trait anxiety and trait fearlessness were not included in the list of PCL-R items, recent research (described in the Anxiety and Fear section) clearly indicates that the current items adequately reflect these two traits (Neumann, Johansson, & Hare, 2013).

The second edition of the PCL-R appeared in 2003, with a greatly expanded manual. The PCL-R items and their scoring criteria remained the same as they had been in the 1991 edition. There are several reasons for the 2003 revision of the manual. First, I sought to minimize the misuse of the PCL-R, especially where it guides or influences adjudication and treatment decisions. One of the most important requirements for proper use of the PCL-R is familiarity with the current literature. However, it was apparent that many of those who produced psychological reports for the criminal justice system or who testified in court relied primarily on material published in the 1991 manual. The inclusion of more recent research in the 2003 manual provided users with additional information about the PCL-R assessment of psychopathy. Nonetheless, it is still extremely important for users to keep abreast of the current literature about psychopathy, especially concerning its implications for minority and legal issues. An up-to-date list of references is available at http://www.hare.org/references.

The second reason for the revision was the large amount of empirical data accumulated since publication of the original manual. The explosion in research provided a large amount of data for establishing comparison tables and for addressing issues concerning the factor structure, reliability, validity, and generalizability of the PCL-R. Psychometric analyses were conducted on 10,896 North American and European offenders and forensic patients, and were described in detail in the manual (Hare, 2003).

Extensive descriptions of the psychometric properties of the PCL-R are available (e.g., Bolt, Hare, Vitale, & Newman, 2004; Forth, Bo, & Kongerslev, 2013; Hare, 2003; Hare et al., 2013; Hare et al., 2018; Mokros et al., 2015), and I provide only a brief outline here.

Administration of the PCL-R

The 2003 Manual for the PCL-R contains a detailed description and scoring instructions for scoring each of its 20 items. It is important for those who use the PCL-R for research or applied purposes to be familiar with the scoring criteria, and not to score an item based on

its title or on what they think it should mean. Besides the Manual, a controlled psychological instrument, several guides for using the PCL-R and its derivatives are available. The edited volume by Archer and Wheeler (2013) contains chapters that discuss in considerable detail the uses and potential abuses of the PCL-R (Hare et al., 2013) and the PCL: YV (Book, Forth, & Clark, 2013). In addition, the volume contains chapters on the forensic use of various other psychological instruments. The edited volume by Gacono (2016) is dedicated almost exclusively to the clinical and forensic use of the PCL instruments, and provides clinicians with sound advice on the use of these instruments in a wide array of areas.

Briefly, the PCL-R is a 20-item construct rating scale widely used in research, clinical, and forensic settings for the assessment of psychopathy in adults (see Table 4.1, left panel). The scale items measure personality traits and behaviors related to traditional conceptions of psychopathy, such as impulsivity, lack of empathy, lack of remorse, irresponsibility, and grandiosity. The assessor uses explicit criteria to rate each item on a 3-point scale (0, 1, or 2) according to the extent to which item description matches the individual. The total score varies from 0 to 40, reflecting the degree to which the individual matches the "prototypical" psychopath.

The standard procedure for administration of the PCL-R involves a semi-structured interview and a review of available file and collateral information (Hare, 2003). The interview typically covers education, employment and family background, relationships, substance use, and antisocial behaviors from adolescence on. It may consist of several sessions, allowing for a more representative sample of the individual's interactional style. The interview itself has several purposes. First, certain historical information about the individual is required for scoring PCL-R items. Second, the interview often provides clues as to the individual's interpersonal style. It also gives the assessor a chance to evaluate consistency (and deception) within the interview and between the interview and collateral information. Third, the use of an interview allows the user to challenge the individual, probe for further information, and resolve inconsistencies. By conducting the file review prior to the interview, the user may have a firm basis for challenging or probing the individual. Finally, it is important to keep in mind that the defining features of psychopathy include manipulation and deception. Harris, Rice, and Cormier (2013) argue that psychopathic individuals engage in positive impression management during the interview, thus resulting in a lowered PCL-R score for them. Similarly, Gillard and Rogers (2015) reported that male jail detainees with a relatively high Factor 1 score were more effective at using positive impression management to reduce their scores on several risk instruments than were those with a relatively low Factor 1 score.

The second part of the assessment involves reviewing any collateral and/or file information that is available; this is a mandatory aspect of the assessment. The use of collateral information allows the rater to evaluate the truth/reliability of the information given in the interview and to gain access to primary data for scoring several of the PCL-R items. Available collateral information will depend upon the setting in which the assessment is conducted. In correctional settings, files typically contain a variety of official records that may be helpful in scoring the items of the PCL-R, and usually are adequate for collateral review (Hare, 2003). In forensic psychiatric and pretrial settings, raters may have access to official reports, but may also be able to examine reports of interviews with family members, friends, and employers, as well as the results of medical and psychological assessments. When dealing

with non-offenders, it may be difficult to obtain sufficient collateral information, although there may be methods available for accessing some useful information, such as telephone interviews, work records, peer ratings, and so forth. Regardless of the context of the assessment, information should come from as many different sources as possible. If raters are not confident in the adequacy of collateral information, they should omit the item.

Descriptive Statistics for the PCL-R

The descriptive statistics for the PCL-R in the second edition are for large pooled samples of 5408 North American (NA) male offenders, 1246 NA male forensic psychiatric patients, and 1218 NA female offenders, all assessed using interview and file information (standard procedure). The Manual also contains data for 1622 NA male offenders and a smaller sample of 402 NA male forensic psychiatric patients assessed using only file information (no interviews). Statistics also are available in an Appendix for 1117 male offenders from the United Kingdom, 265 Swedish male offenders (file reviews), and 350 Swedish forensic psychiatric patients (file reviews).

The largest difference among NA data sets is the relatively low scores for offenders assessed using only file information. The use of file information alone is a nonstandard procedure but one that researchers use when they are unable to conduct interviews with offenders or patients. In some cases, the file reviews are part of postdictive or retrospective analyses designed to identify variables related to behaviors or events that occurred before or after the compilation of archival data. The advantages of such analyses are that a great deal of useful information can be gathered in a short time, and that it is not necessary to wait years to find out if a variable "predicts" a particular behavior or event (e.g., recidivism). A disadvantage, at least with respect to PCL-R assessments, is that without an interview there may not be sufficient information to score adequately the items that tap interpersonal and affective features. The rater then must rely on what others have recorded in the files.

Reliability of PCL-R Scores

Like its predecessor, the second edition manual provided strong evidence for the reliability of the PCL-R items and Total and Factor scores. Internal consistency was generally high (alpha, mean inter-item correlation), as was inter-rater reliability for single ratings (ICC_1) and for the average of two ratings (ICC_2). For the pooled standard assessment data sets, ICC_1 and ICC_2 values for the total score were 0.87 and 0.93, respectively. For the pooled file review data sets, alpha was 0.87 and the mean inter-item correlation was 0.25. Reliabilities for the UK and Swedish samples were comparable: For the UK sample, coefficient alpha was 0.79, ICC_1 was 0.89, and ICC_2 was 0.94; for the pooled Swedish samples, coefficient alpha was 0.81 (ICCs were unavailable).

A perusal of the literature clearly indicates that researchers and graduate students have no difficulty in obtaining high inter-rater reliabilities for the PCL-R, with ICC_1 for total scores typically being in the 0.85–0.90 range for a given study or laboratory.[1] Similar values

have been reported when comparing research-based ratings with those made by correctional psychologists (e.g., Brown & Forth, 1997; Willemsen, Vanheule, & Verhaeghe, 2011; Woodworth & Porter, 2002).

Scores are reliable in institutional settings when the raters are well trained and careful in their assessments. For example, Sample B-2 in the 2003 Manual described the PCL-R scores of 448 male offenders assessed by at least two independent raters working in the Her Majesty's Prison Service (HMP). Adelle Forth and I trained the interviewers in this sample to a high standard in a series of workshops for HMP. The ICC_1 for this sample was 0.89, while ICC_2 was 0.93.

In the sections on Field Reliability and Adversarial Issues, I discuss the reliability of PCL-R assessments done "in the field."

Standard Error of Measurement

The standard error of measurement is the standard deviation of observed scores if the true score is held constant. The more reliable a measure is and the smaller its standard deviation, the smaller the standard error of measurement. In the 1991 edition of the PCL-R, the estimated standard error of measurement was 3.25 for total scores. As more and more researchers began to use the PCL-R, the reliability of assessments increased. The standard error of measurement in the second edition thus is lower for all comparison samples: NA and UK samples = approximately 3.0 for a single rating and 2.0 for averaged ratings; Swedish sample = 3.0 for a single rating, as estimated from coefficient alpha. The standard error of measurement is an estimate of "the reasonable limits of the true score for [(an individual)] with any given obtained score" (Anastasi, 1982).

PCL-R Thresholds for Psychopathy

There is good evidence that at the measurement level the structure of psychopathy is dimensional as assessed with the PCL-R, its derivatives, and various self-report measures of psychopathy (Edens, Skeem, & Douglas, 2006; Guay, Knight, Ruscio, & Hare, 2018; Guay, Ruscio, Knight, & Hare, 2007; Murrie et al., 2007; Walters, Ermer, Knight, & Kiehl, 2015). Nonetheless, for *research*, it may be convenient to adopt a particular threshold for psychopathy, such as a PCL-R score of 30 (out of 40). Investigators in some European countries use a score of 25 or 26. The use of different threshold scores reflects differences in the distribution of psychopathy scores. However, this does not mean that the scores are metrically equivalent to one another (see Bolt, Hare, & Neumann, 2007; Bolt et al., 2004). Nonetheless, the properties of the PCL-R and its correlates have considerable cross-cultural generalizability. Thus far, there are some 20 translations of the PCL-R, with perhaps a dozen or so unauthorized translations in use, with unknown properties.

Problems arise when clinicians and investigators treat a cut score as if it were a sharp (or even fuzzy) dividing line between offenders and patients who are psychopaths (members of a taxon) and those who are not. The problem is more severe when the judicial system makes the same assumption (Hare, 1998b). We view thresholds primarily as a methodological

convenience for facilitating communication among researchers; they are not gatekeepers for a putative taxon. Their utility for clinical and forensic purposes will depend on the context in which the PCL-R is used (e.g., diagnosis, risk assessment, treatment options, and so forth), as well as an appreciation of measurement error and the issues associated with the use of cut scores (Brenner & Gefeller, 1997; Hare, 2003). Many investigators prefer to use methods that do not rely on a set cut score and that use all the information available, such as correlational analyses, multiple regressions, and structural equation modeling (SEM; Neumann, Kosson, & Salekin, 2007).

The above discussion notwithstanding, the taxonic/dimensionality issue with respect to psychopathy is not closed. Latent profile analysis (LPA) has identified a latent class (C1) with very high scores on all four PCL-R facets (Mokros, Hare, Neumann, & Habermeyer, Chapter 5 this volume; Neumann, Vitacco, & Mokros, 2016). The following quote from Ruscio (2007, p. 1589) is informative in this regard:

> For example, psychopaths might differ qualitatively from nonpsychopaths; psychopathy may be taxonic. Alternatively, individuals might differ from one another along one or more continuous traits; psychopathy may be dimensional. *It is possible that the structure of psychopathy may be more complex, containing both taxonic and dimensional features:* Within a nonpsychopathic group, a set of dimensions may capture individual differences on psychopathy-relevant traits, but hardcore psychopaths deviate substantially on multiple dimensions and form their own group (emphasis mine).

Factor Structure of the PCL-R

Early factor analyses of the PCL indicated that the items fell into two broad clusters or factors (Harpur et al., 1988; Harpur et al., 1989). Factor 1 reflected the interpersonal and affective components of the construct, whereas Factor 2 was more akin to a socially deviant (externalizing) lifestyle. This two-factor model had a large influence on the conceptualization of psychopathy and empirical research over the next decade or so.

Hare et al. (1990) obtained a similar two-factor solution in a factor analysis of the PCL-R items of 1281 male offenders and forensic psychiatric patients. The sample was independent of those used within the PCL analysis. The manual for the 2003 edition of the PCL-R contains large data sets for NA male and female offenders and male forensic psychiatric patients. Exploratory and confirmatory factor analyses by Neumann, Hare, and Newman (2007) of the male offender data set yielded support for a superordinate factor of psychopathy underpinned by four correlated first-order factors, referred to here as facets: *Interpersonal*, *Affective*, *Lifestyle*, and *Antisocial* (Table 4.1 left panel; Figure 4.1). Subsequent confirmatory factor analysis (CFA) of the pooled data for male and female offenders and for male forensic psychiatric patients (Neumann, Kosson, et al., 2007) indicated that the four-correlated-factor model fit the data well.

Using the above NA data sets, Hare and Neumann (2008) obtained evidence for an alternative, compatible model specifying two higher-order factors (corresponding to the original Factors 1 and 2, with item 20 added to the latter), each encompassing two lower-order

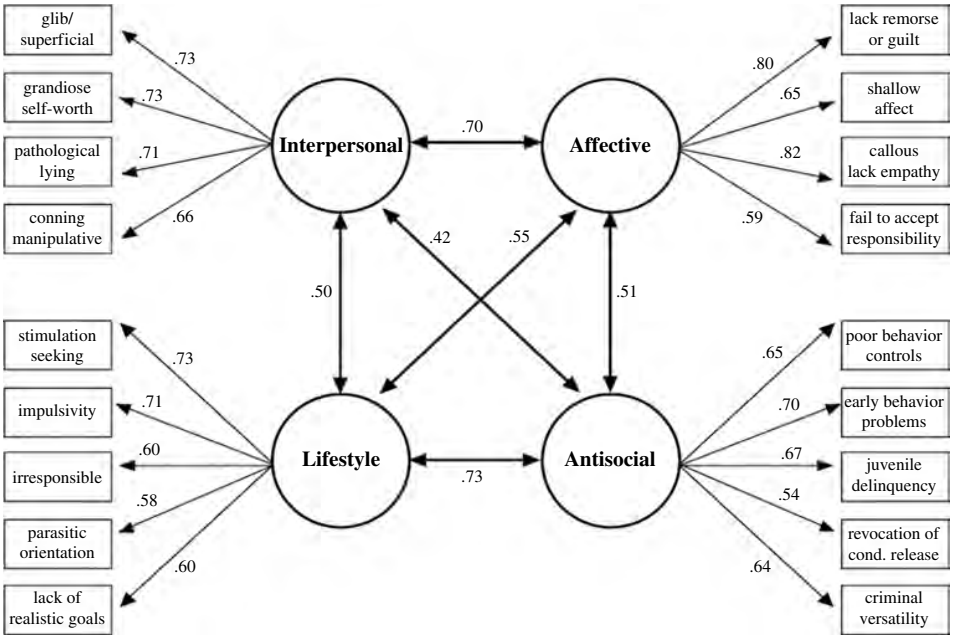


Figure 4.1 Four-factor PCL-R item-based model of psychopathy for male and female offenders and male forensic psychiatric patients (*N* = 6929). *Source:* From Neumann, Hare, & Neumann (2007). Reprinted with permission of Guilford Press.

factors—with the lower-order facets as a whole corresponding to those of the four-factor model. Fit statistics for this model, depicted in Figure 4.2, matched those for the basic four-factor model in this same sample. In these analyses, we used Mplus (Muthén & Muthén, 2001) for the CFA, treated the items as ordinal variables, and used a robust weighted least-squares parameter estimation procedure. This model has proved robust across many different samples from North America and Europe. For example, Neumann, Vitacco, and Mokros (2016; see also Hare, 2016) conducted a CFA on the PCL-R items of a mega-sample (N = 12 301) consisting of 6929 NA male offenders and 5372 European male offenders. Statistical model fit for the four-factor model "was good, indicating that it has excellent generalizability across cultures" (p. 18). In a larger review, Neumann et al. (2015) summarized the evidence for a four-factor model of the PCL-R in samples of North American, South American, English, and European male offenders (N = 13 242), and of North American female offenders (N = 1099). Statistical fit generally was good across samples. In a recent article, Eisenbarth, Krammer, Edwards, Kiehl, and Neumann (2018) found that the four-factor model was applicable in a sample of 155 female Hispanic offenders. The available evidence indicates that the PCL-R has much the same psychometric properties, structure, and correlates in other countries as in North America (e.g., Fanti, Lordos, Sullivan, & Kosson, 2018; Mokros, Habermeyer, et al., 2014; Mokros et al., 2011; Zwets, Hornsveld, Neumann, Muris, & van Marle, 2015). However, item response theory (IRT) analyses suggest that there may be ethnic and cultural differences in the way some features of psychopathy are manifested (e.g., Bolt et al., 2004; Bolt et al., 2007; Cooke & Michie,

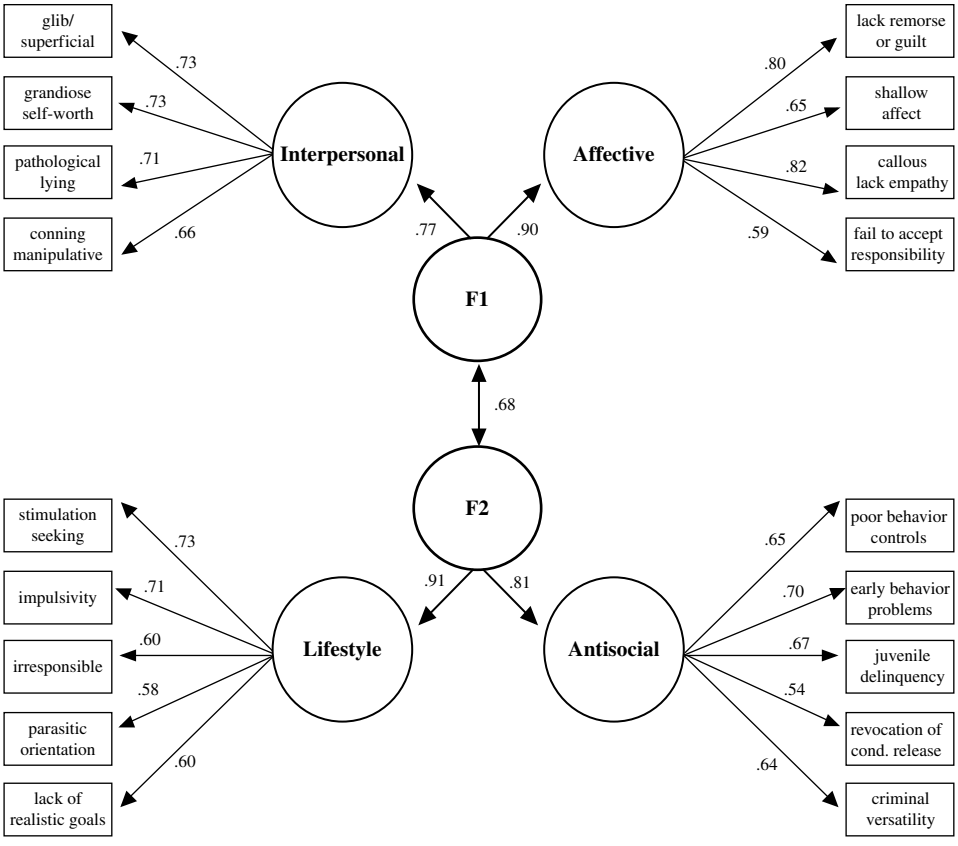


Figure 4.2 Two-factor PCL-R higher-order representation of the four correlated factors model (*N* = 6929). *Source:* From Hare & Neumann (2008).

1997; Cooke, Michie, Hart, & Hare, 1999; Hare et al., 2018). The broader implication is that cultural factors may affect which items, factors, and facets are most discriminating of psychopathy in a given context (Fanti et al., 2018).

Antisociality

Some investigators have argued that the Antisocial facet, or more generally, antisociality, is not part of the psychopathy construct (e.g., Cooke & Michie, 2001; Skeem & Cooke, 2010). However, there is ample evidence to conclude that antisociality is a fundamental part of the construct (e.g., Hare & Neumann, 2010; Miller & Lynam, 2015a; Neumann et al., 2015). As put by Lynam and Miller (2012): "The most ubiquitous characteristic in the descriptions of psychopathy by Cleckley, Hare, Karpman, Kraepelin, Lykken, and the McCords is a history of severe, longstanding ASB [antisocial behaviour]" (p. 346). Miller and Lynam (2015b) listed the recognition of the role of antisocial behavior as one of the five important advances in theory and research. They note that in the absence of antisociality, "psychopathy

becomes a configuration of traits that is interesting to look at but that has little real world consequence, reducing psychopathy to a sort of boutique personality disorder." Empirical evidence that antisociality is an integral part of the psychopathy construct is extensive; it is clear that it properly belongs in the four-factor model of the PCL scales (Neumann, Hare et al., 2007; Neumann, Vitacco, et al., 2016). Neumann et al. (2015) obtained good fit for the four-factor model in a large mega-sample of 52,957 individuals with a variety of PCL-based scales from different samples (male and female offenders and psychiatric patients, adolescent male and female offenders, and community) and countries.

The four-factor model of the PCL-R and its derivatives is proving to be very useful for delineating the dimensions of the psychopathy construct and for clarifying the differential associations of the facets with many external variables having to do with etiology, developmental course, risk evaluations, treatment options, cognitive/affective neuroscience, and so forth (e.g., Gacono, 2016; Kiehl & Sinnott-Armstrong, 2013; Patrick, 2018). SEM relating these facets to outcome variables is proving to be particularly informative about the nature and correlates of psychopathy (Neumann, Vitacco, et al., 2016).

The factor structure of the PCL-R (and its derivatives; see below) is of more than academic interest. For example, clinicians may describe an individual offender, patient, or client in terms of the score on each of the four facets, thus developing a profile that helps to provide a more finely nuanced picture of the individual. These variable-centered approaches provide the basis for person-centered approaches (Neumann, Vitacco, et al., 2016). LPA uncovers patterns of facet scores, or latent classes that have strong explanatory and predictive power. Details about this new research are available elsewhere (Hare, 2016; Mokros et al., Chapter 5 this volume; Neumann, Vitacco, et al., 2016).

There are several alternative structural models of the PCL-R. Cooke and Michie (2001) developed a three-factor model based on 13 items from the Interpersonal, Affective, and Lifestyle dimensions. Patrick, Hicks, Nichol, and Krueger (2007) constructed a *bifactor* model "that included a General factor on which all items of the PCL-R loaded significantly, along with three separate subfactors [Interpersonal, Affective, and Impulsivity] on which particular subsets of PCL-R items loaded" (p. 135). The common factor in a bifactor model of the PCL-R would reflect the practice of summing all item ratings into a single total score. The connection with the four-factor model is that in the bifactor model the four facets are nested nuisance factors that capture specific variance not accounted for by the common factor. As a bifactor model (with one general factor and four such nuisance factors) is nested within a four-factor model involving a superordinate psychopathy factor (cf., Neumann, Hare, et al., 2007), the two models can be compared with each other directly in terms of relative model fit. Hare et al. (2018) discuss the three-factor and bifactor models in some detail. Bonifay, Lane, and Reise (2017) have outlined concerns about the use of a bifactor model in psychopathology. Recently, Walters (2015) presented a two-factor model (labeled Fearlessness and Disinhibition) based on 10 items selected from the Interpersonal, Affective, and Lifestyle dimensions of the PCL-R. As noted elsewhere (Hare et al., 2018, p. 69), Fearlessness is a misnomer for a factor defined by three items from the PCL-R's Interpersonal facet (Glib/superficial, Pathological lying, and Conning/manipulative) and three items from its Affective facet (Lack of remorse, Shallow affect, and Callous/lacks empathy). Presumably, Walters referred to this set of items as Fearlessness to make it appear that parts of the PCL-R were in line with the TriPM dimension of Boldness and the Psychopathic Personality Inventory (PPI) dimension of Fearless Dominance (FD).

Anxiety and Fear

As noted above, some commentators have expressed concern that the PCL instruments do not contain items for Low Anxiety and Fearlessness. The reasons for not including them are described in detail elsewhere (Hare, 2003, Hare & Neumann, 2008).[2] One reason was that it was apparent early in the development of the PCL that specific measures of negative affect would add little to what the other items already covered. A study by Neumann, Johansson, and Hare (2013) illustrates this point. The authors had access to a PCL-R data set for 398 Swedish violent offenders, along with items designed to index low anxiety and fearlessness, and cast in the form of the PCL-R item protocol. The *Low Anxiety* item covered cognitive, emotional, and behavioral features associated with a relative absence of anxiousness. The *Fearlessness* item pertained to engagement in a variety of risky behaviors, with little evidence of subjective fear. Clinicians scored these items on a 3-point scale (mirroring the PCL-R scoring format) with information from a semi-structured interview and file records. CFA of the PCL-R items replicated the four-factor model. Additional CFAs incorporating the Low Anxiety and Fearlessness items indicated that these items, either individually or as a pair, could be on any of the PCL-R facets without any change in model fit. These results indicated that the extant PCL-R item set reflects tendencies toward low anxiety and fearlessness. The authors concluded that there should be no concerns about the absence of specific items for low anxiety and fearlessness in the PCL-R, and that the PCL-R items reflect a general attenuation of affective experience.

Validity of the PCL-R

Evidence for the validity of the PCL-R and the construct it measures is extensive and the literature too voluminous to even summarize here. Suffice it to say that the evidence comes from many diverse areas of psychology, psychiatry, biology, sociology, criminology, behavior genetics, developmental psychopathology, the criminal justice and legal systems, and affective/cognitive neuroscience. Much of this evidence is described in detail elsewhere (e.g., Gacono, 2016; Häkkänen-Nyholm & Nyholm, 2012; Hare, 2003; Hare et al., 2013; Hervé &Yuille, 2007; Kiehl & Sinnott-Armstrong, 2013; Patrick, 2006, 2018; Singh et al., 2016). Many of the chapters in the two volumes of this handbook provide evidence for the validity of the PCL-R in a variety of contexts. For an up-to-date listing of articles, chapters, and books on psychopathy, see www.hare.org.

Derivative Measures

The primary use of the PCL-R is with criminal and forensic psychiatric populations, with occasional extensions to the general community. Part of its strength is the requirement that in order to score an item, the rater has to supplement an in-depth interview with extensive collateral information. During the planning phase in the late 1980s of what was to become the MacArthur Study of Mental Disorder and Violence in civil psychiatric patients (see Monahan et al., 2001), it became evident that a potentially important risk factor for violence

was psychopathy. The study was in need of a version of the PCL-R scale that required less time, effort, and information for a reliable assessment. The result was the PCL:SV, which was the single most important risk factor for violence in the study population (Steadman et al., 1999).

PCL:SV

The PCL:SV consists of 12 items derived from the PCL-R, each scored on a 3-point scale (0, 1, 2) based on interview and collateral information that is less extensive than that required for scoring the PCL-R. Total scores can vary from 0 to 24. It is related, conceptually and empirically, to the PCL-R (Cooke et al., 1999; Hart et al., 1995). Although it may function a screen for psychopathy in forensic populations (Guy & Douglas, 2006), it is a stand-alone instrument for research with noncriminals, including civil psychiatric patients (as in the MacArthur study). Its psychometric properties and four-factor structure (see Table 4.1, right panel) are much the same as those of the PCL-R (Hill, Neumann, & Rogers, 2004; Vitacco, Neumann, & Jackson, 2005; Žukauskienė, Laurinavičius, & Čėsnienė, 2010). There is rapidly accumulating evidence for the construct validity of the PCL:SV, including its ability to predict aggression and violence in offenders and in forensic and civil psychiatric patients. In this respect, the correlates of the PCL:SV are much the same as those of the PCL-R.

PCL:YV

The PCL:YV (Forth et al., 2003) is an age-appropriate modification of the PCL-R intended for use with adolescents. Like the PCL-R, it consists of 20 items and four factors (Jones, Cauffman, Miller, & Mulvey, 2006; Kosson, Cyterski, Steuerwald, Neumann, & Walker-Matthews, 2002; Kosson et al., 2013; Neumann, Kosson, Forth, & Hare, 2006). Table 4.1 (middle panel) lists the items. It has much the same psychometric properties and much the same correlates as its adult counterpart, both male and female (Anderson & Kiehl, 2014; Book et al., 2013; Kosson et al., 2013; Salekin, Neumann, Leistico, DiCiccio, & Duros, 2004; Vitacco, Neumann, Caldwell, Leistico, & Van Rybroek, 2006; Walters, & Kiehl, 2015).

Like the PCL-R, the PCL:YV appears to generalize well across ethnic groups and countries (e.g., Book et al., 2013; Dolan & Rennie, 2006; Hillege, de Ruiter, Smits, van der Baan, & Das, 2011; McCoy & Edens, 2006; Schrum & Salekin, 2006; Tsang et al., 2015). Although there is little doubt about the reliability and validity of the PCL:YV, concerns arise with respect to its use in the criminal justice system. The main issues have to do with (a) the dangers of labeling an adolescent as a psychopath; (b) the implications of the PCL:YV for classification, sentencing, and treatment; (c) the possibility that some features measured by the PCL:YV are found in normally developing youth; and (d) the degree of stability of psychopathy-related traits from late childhood to early adulthood. Extensive discussions of these issues are available elsewhere (e.g., Book et al., 2013; Forth & Book, 2007; Hawes et al., 2018; Lynam & Gudonis, 2005; Frick, Ray, Thornton, & Kahn, 2014; Salekin, 2016; Salekin & Lynam, 2010; Vitacco & Vincent, 2006). Briefly, although psychopathy and its features do not suddenly emerge in early adulthood, it would be inappropriate to label an

adolescent as psychopathic or to use a high PCL:YV score as a basis for a harsher sentence or for exclusion from treatment. Although some adolescents may exhibit some features of psychopathy in certain contexts or for a limited time, a high score on the PCL:YV requires evidence that the traits and behaviors are extreme and manifest themselves across social contexts and over substantial times. High ratings of psychopathic traits are rare in community youth (Book et al., 2013). As Lynam & Gudonis (2005) put it following their review of the literature:

> [P]sychopathy in juveniles looks much like psychopathy in adults. The same traits characterize these individuals at different developmental time points. Additionally, juvenile psychopathy acts like adult psychopathy. Like their adult counterparts, psychopathic juveniles are serious and stable offenders. They are prone to externalizing disorders .... as far as has been observed, juvenile psychopathy appears quite stable across adolescence. All of these findings replicate those observed in studies using psychopathic adults. (pp. 401–402)

Antisocial Process Screening Device (APSD)

The APSD (Frick & Hare, 2001) is a teacher/parent rating scale for use with children from 6 to 13 years of age. It consists of 20 items that measure three dimensions of behavior thought to be precursors to psychopathic traits: Callous/unemotional (CU), Narcissism, and Impulsivity. Recently, the research emphasis has been on the CU traits, with considerable evidence that they are heritable, related to a variety of disruptive behaviors, and relatively stable over three- or four-year periods (Frick & Ray, 2015; Frick, Wall, Barry, & Bodin, 2016; Viding & McCrory, 2012). An expanded version of the CU traits measured by the APSD is available as The Inventory of Callous-Unemotional Traits (Essau, Sasagawa, & Frick, 2006; Roose, Bijttbier, Decoene, Claes, & Frick, 2010). As indicated by Frick et al. (2014, p. 3):

> ... in the past two decades, a significant body of research has emerged refining how the key features associated with psychopathy may be expressed in children and adolescents (Frick, 2009; Salekin, 2006).These conceptualizations have focused largely on the presence of callous-unemotional (CU) traits (e.g., lack of empathy and guilt, failure to put for the effort on important tasks, shallow and deficient emotions), which correspond closely to the affective dimension of psychopathy—core to the construct in adult samples. (Hare & Neumann, 2008)

These authors provide an extensive review of the research on CU traits and on their role as a specifier for Conduct Disorder in DSM-5.

In commenting on the use of CU traits as a conduct disorder specifier, Salekin (2016) reviewed the relevant literature and argued that CU traits alone do not capture the essence of psychopathy-related traits in childhood: "By using seminal psychopathy models (Cleckley, 1941; Hare, 1991/2003), one can begin to determine the degree of overlap, for example, from key conceptualizations of psychopathy to the ICD-10 and the newly devised

DSM–5 CD criteria" (Salekin, 2016, pp. 180–181). He proposed that conduct disorder should have three specifiers: Grandiose-Manipulative, Callous-Unemotional, and Daring-Impulsive. The specifiers correspond, respectively, to the Interpersonal, Affective, and Lifestyle facets of the PCL-R/PCL: YV. In a personal communication to the author (May 5, 2016), Salekin had this to say:

> Child psychopathy research has consistently shown to be represented by a three-factor model including Grandiose-Manipulative traits, CU traits, and Daring-Impulsive traits, and when including CD, you essentially have the four-factor "model of psychopathy." This fits well with Quay's [1987] notion that psychopathic traits can inform CD, and CD actually helps ensure that some conduct problems are present. Similarly, this childhood model might also help re-inform adult research, which has shown that a four factor model is statistically viable and particularly meaningful in identifying this severe personality condition.

The issues raised by Salekin have generated conceptual and empirical interest, and are the subject of several articles in a Special Section on Childhood Psychopathic Traits for Specifying Conduct Order (Salekin, Andershed, Batky, & Bontemps, 2018). In reviewing the articles in the Special Issue, Lee (2018) commented:

> The innovations reflected in these important contributions should catalyze deeper thinking and more refined efforts at understanding the origins, development, and nature of youth psychopathic traits. Their collective consistency in implicating multiple dimensions of youth psychopathic traits as predictors of important outcomes challenges assumptions around the primacy of CU traits.

Research along these lines is well under way. Salekin and Hare (2016) developed the *Proposed Specifiers for Conduct Disorder* (PSCD). Recent research with a variety of different cultures supports the utility of the model and its rationale (e.g. Lopez-Romero et al., 2019; Luo, Wang, Neumann, Hare, & Salekin, 2019).

Childhood Psychopathy Scale (CPS)

Like the APSD, the CPS (Lynam, 1997) is "a downward developmental translation of what is arguably the 'gold standard' for the assessment of psychopathy in adulthood: the Hare Psychopathy Checklist—Revised" (Lynam, Derefinko, Caspi, Loeber, & Stouthamer-Loeber, 2007, p. 363). The development of the CPS, a parent-teacher rating scale, is described by Lynam (1997). It consists of 12 items (each with 2–4 descriptors) operationalized from descriptions of PCL-R items: glibness, untruthfulness, manipulation, lack of guilt, poverty of affect, callousness, parasitic lifestyle, behavioral dyscontrol, lack of planning, impulsiveness, unreliability, and failure to accept responsibility. The CPS is a reliable and valid indicator of psychopathic traits in children and adolescents (Bijttebier, & Decoene, 2009; Breaux et al., 2019; Lynam et al., 2007) and has proved to be very useful in helping to track the longitudinal psychopathy-related features from childhood through to adulthood. For example, Hawes et al. (2018), using the CPS with in a large cohort of youth ($N = 1517$), reported:

> Psychopathy features assessed in this study exhibited measurement consistency across development (ages 7–16) and were found to tap into the same underlying construct as intended across measurement occasions, and equivalently for African-American and Caucasian youth. Results were similar when assessed using traditional IRT procedures for longitudinal invariance testing and when implementing the more recent [Bayesian structural equation modeling] BSEM methodology.

Business Scan (B-Scan)

The B-Scan is a measure of psychopathy-related traits and behaviors designed for use in business-oriented or organizational settings (Mathieu, Hare, Jones, Babiak, & Neumann, 2013). It comprises items related to work behaviors, written in a business-friendly, non-clinical language, and intended to engage those with a corporate mindset. The B-Scan has two versions: a self-report version (B-Scan Self) and a measure of other's psychopathy traits (B-Scan 360). So far, validation studies on the B-Scan have identified four factors, which correlate with Hare's four-factor model of psychopathy (Mathieu et al., 2013; Mathieu, Neumann, Babiak, & Hare, 2015). The B-Scan requires more validation before it is suitable for use within corporate settings. For discussions of psychopathy in the workplace, see Babiak and Hare (2019), and Mathieu, Babiak, and Hare (Chapter 27 this volume).

ICD-10 and DSM-5

The attributes measured by the PCL-R and its derivatives are similar to but not identical with the diagnostic criteria for dissocial personality disorder listed in ICD-10 (Huchzermeier et al., 2007; Ullrich & Marneros 2004, 2007; Widiger et al., 1996), and the criteria for ASPD in DSM-III (American Psychiatric Association, 1980), DSM-IV (American Psychiatric Association, 1994), and DSM-5 (American Psychiatric Association, 2013). There is a particularly large literature on the associations of the PCL scales with the DSM-III and DSM-IV diagnostic category of ASPD. Several recent historical accounts of these associations are available, indicating that they are not the same clinical constructs (Crego & Widiger, 2015; Hare et al., 2012; Lykken, 2018; Lynam & Vachon, 2012;Warren & Burnette, 2013; Widiger, 2018), and I summarize only a few salient details here.

In developing the DSM-III criteria for ASPD, the intention was to make them compatible with traditional conceptions of psychopathy (Crego & Widiger, 2015; Hare, 2003). However, the criteria selected for ASPD reflected observable, mostly antisocial, behaviors. The criteria remained the same in DSM-IV. It has long been known that there is an asymmetric association between the PCL-R and ASPD; most offenders with a high PCL-R score meet the criteria for ASPD, but most of those diagnosed with ASPD do not have high PCL-R scores (Hare, 2003; Ogloff, 2006; Patrick, 2007; Warren & Burnette, 2013). In this respect, it is noteworthy that ASPD is strongly associated with PCL-R Factor 2 items but only weakly associated with Factor 1 items, leading to the concern that essential elements of psychopathy were not included in the criteria for ASPD. This concern also applies to ASPD as defined in DSM-III, DSM-IV, and Section II of DSM-5.

After years of debate about the need to bring the diagnostic criteria for ASPD into line with those for psychopathy, DSM-5 retained the DSM-IV criteria for ASPD (Section II, Diagnostic Criteria and Codes). Many clinicians and researchers saw this as a surprising development, given earlier suggestions and proposals by the Personality Disorders (PDs) Work Group for DSM-5. About this issue, Crego and Widiger (2015 p. 670) had this to say:

> Ever since DSM-III was published, there has been a recurrent criticism of the American Psychiatric Association (APA) diagnostic manual for failing to be fully commensurate with the conceptualization of psychopathy by Cleckley (1941, 1976) and/or the PCL(−R). (Hare, 1980, 2003)

Crego and Widiger (2015, p. 668) also noted:

> It was the intention of the authors of the DSM-IV ASPD to shift the diagnosis closer still to the PCL-R conceptualization (Widiger & Corbitt, 1993). Considered for DSM-IV was an abbreviated version of the PCL-R developed by Dr. Hare, consisting of 10 items. (Hare, 2003; Widiger & Corbitt, 1993)

Crego and Widiger (2015) suggested that "the authors of DSM-5, however, may have shifted the ASPD toward their own, more recently developed, conceptualization of psychopathy" (p. 670).

In early 2010, the DSM-5 Work Group proposed that ASPD be renamed Antisocial/Psychopathic Type, reflecting traits from domains of Antagonism and Disinhibition, ostensibly similar to PCL-R Factors 1 and 2, respectively. Samuel, Lynam, Widiger, and Ball (2012) had 170 experts perform prototypicality ratings of the traits proposed for ASPD, as well as those for five other proposed personality disorders. They reported that the "Work Group and the experts appeared to be in complete agreement about the traits that should and should not be used for the antisocial/psychopathic type" (p. 6).

Skodol et al. (2011, p. 140) stated that "...a revised construct of ASPD that includes psychopathic personality features has been recommended for retention in DSM-5." Yet, in their rational for this proposal, they did not refer to the enormous body of research with the PCL scales. As Blashfield and Reynolds (2012) noted, "Cleckley and Hare are well-known authors who defined how psychopathy is currently conceptualized; neither was referenced in the DSM-5 rationale" (p. 826). Lynam and Vachon (2012) commented that "... history and research suggest that ASPD and psychopathy should be combined in DSM-5" (p. 490), but that the issue apparently received little or no discussion in the literature or on the DSM-5 websites. Further, "... the DSM Workgroup missed an opportunity to unify two classifications (ASPD and psychopathy) that history and research suggest have diverged mistakenly" (p. 492).

However, the door for some integration remains slightly ajar in the dimensional-trait characterization of ASPD in Section III of DSM-5, which describes ASPD in terms of the traits of manipulativeness, deceitfulness, callousness, hostility, irresponsibility, impulsivity, and risk-taking from the broad domains of Antagonism and Disinhibition. Anderson, Sellbom, Wygant, Salekin, and Krueger (2014) suggested that use of these traits has the "potential to move the diagnosis of ASPD closer to the more useful construct of psychopathy and also

allow for more flexibility in characterizing individuals with psychopathic personality traits" (p. 676). These traits are included in the Personality Inventory for DSM-5 (PID-5; Krueger, Derringer, Markon, Watson, & Skodol, 2012), designed to operationalize the DSM-5 Section III dimensional-trait model in the domain of self-report. Strickland, Drislane, Lucy, Krueger, and Patrick (2013) noted that self-reports are useful "for indexing antisocial/psychopathic tendencies and personality more broadly," but that corroboration of their findings will require additional research "using alternative measurement methods such as face-to-face interview supplemented by archival file review" (p. 336).

DSM-5 also includes a psychopathy specifier for the trait-based diagnosis of ASPD. It includes three additional traits: low anxiousness, high attention-seeking, and low social withdrawal. These are indicators of Boldness in the Triarchic Psychopathy Measure (TriPM; Patrick & Drislane, 2015), and Fearless Dominance in the Psychopathic Personality Inventory-Revised (PPI-R; Lilienfeld & Widows, 2005). Anderson et al. (2014) reported that in university and community samples, the PID-5 and the psychopathy specifier correlate with the TriPM and the PPI, referred to as "extant conceptualizations of the psychopathy construct" (p. 690). However, the TriPM and PPI-R diverge in some ways from the traditional construct of psychopathy (Crego & Widiger, 2015; Hare et al., 2012; Lynam & Miller, 2012; Miller & Lynam, 2012; Neumann, Uzieblo, Crombez, & Hare, 2013), making it important to determine how the psychopathy specifier relates to the PCL scales. Few, Lynam, Maples, MacKillop, and Miller (2015) noted that "the inclusion in the DSM-5 of the psychopathy specifier was somewhat surprising as there is little research examining its validity when measured in this way..." (p. 72). These authors reported that the DSM-5 psychopathy specifier traits were not significantly related to some measures of psychopathy (Self-Report Psychopathy [SRP], Five Factor Model [FFM]) or externalizing scores, but were related to FD, which they argue is not part of the psychopathy construct (see Patrick & Drislane, 2015 for an alternative perspective). It remains to be seen if Section III of DSM-5 will facilitate our understanding and measurement of psychopathy, or prove useful in clinical, forensic, and other applied applications where assessments have serious implications for the individual and society.

Self-Report Measures of Psychopathy

The clinical and empirical interest in psychopathy generated by the PCL-R prompted development of several self-report measures of psychopathy. In addition to the PPI-R and the TriPM, mentioned above, there are the Levenson Self-Report Psychopathy scale (LSRP; Levenson, Kiehl, & Fitzpatrick, 1995), the Youth Psychopathic Traits Inventory (YPI; Andershed, Kerr, Stattin & Levander, 2002); the Elemental Psychopathy Assessment (EPA; Lynam et al., 2011; Lynam & Miller, 2015); the Antisocial Behavior (RC4) scale from the Minnesota Multiphasic Personality Inventory-2-RF (Ben-Porath, 2012); and the Self-Report Psychopathy-4 (SRP-4) scale (Paulhus, Neumann, & Hare, 2016). The EPA views psychopathy in terms of basic personality traits ("from the ground up"). However, it originated in translation of the PCL-R item criteria into the language of the FFM of personality (Widiger & Lynam, 1998).The SRP-4 is directly tied to the PCL-R and has the same four-factor structure.

There are several advantages and disadvantages to using specialized and general scales of this sort for the study of psychopathy (Miller & Lynam, 2015b; Sellbom, Lilienfeld, McCrary, & Fowler, 2018; Westen & Weinberger, 2004). They make it possible to use very large samples, in both forensic and other populations. They broaden the nomological network by extending to the general population some of the research paradigms found useful in experimental/laboratory research with forensic populations. Further, by viewing psychopathy from different perspectives, including that of the individual under study (self-reports), these scales may lead to a more complete and accurate conceptualization of the construct when combined with other measures (Sellbom et al., 2018). Moreover, they make the researcher's task much easier, because the study participants, who provide their own managed view of themselves, do most of the assessment work. Neumann, Schmitt, Carter, Embley, and Hare (2012) have provided a discussion of the role of cultural factors in self-reported psychopathic traits.

However, we should recognize that psychopathic individuals are particularly prone to impression management (Cooper & Yuille, 2007). Kelsey, Rogers, and Robinson (2015) noted that "low scores on self-report measures of psychopathy may indicate genuinely low psychopathy or they may reflect intentionally suppressed moderate to high levels of psychopathy" (p. 390). Ray, Hall, Rivera-Hudson, Poythress, Lilienfeld, and Morano (2013) reported that the effect of response bias on self-reports (the PPI, PPI-R, and the LSRP) were not large, but also concluded that self-report psychopathy measures cannot "safely replace more extensive clinical assessments, such as the PCL-R" (p. 13). In my view, self-report scales are very useful for community and forensic research in which the respondents are anonymous, perhaps paid for their participation, and the results have no implications for the participants. The situation is much different for the clinical and forensic assessment of individuals, where the stakes are very high, both for the individuals and society. If an offender completes a self-report knowing that the decision-makers will see the results, it is likely that the responses will be very much different from those of an offender who takes part in research that maintains anonymity.

We should ensure that we do not cast the nomological net too wide or in the wrong places, and that the construct putatively under investigation really is what we think it is. For example, in a commentary on the meaning and utility of the psychopathy concept among adolescents, Rutter (2005, p. 500) discussed the view held by some (e.g., Lynam et al., 2005) that we can conceptualize psychopathy as a particular pattern of normal personality dimensions, including emotional detachment. Rutter questioned whether or not individuals who do not appreciate the feelings of others can provide valid reports about their own emotions.

> On the other hand, at least as originally put forward by Cleckley (1941), psychopathy was supposed to represent something that was a bit different from normal variations in personality. That is, of course, not to argue that psychopathy is a categorical, rather than a dimensional, feature but it does suggest that the characteristic of emotional detachment may not be best represented by scores on personality dimensions that are designed to pick up rather different features. (Rutter, 2005, p. 500)

The use of different methods for measuring psychopathy is predicated on the assumption that they all identify much (or parts of) the same thing. However, Babiak and Hare (2019,

p. 28) noted that "it also is possible that they represent conceptualizations of different or weakly related constructs, albeit using the same name (the "jingle fallacy"). For this reason, it should be incumbent for authors to include the name of their measurement tool in the Abstracts of their articles."

Field Reliability[3]

Because of its prominence as an international standard for the clinical/forensic/research assessment of psychopathy, and in recognition of the important role it plays in the criminal justice system, the PCL-R is the subject of unusually intense scrutiny and critical analyses, both conceptual and statistical. It has fared well, not only with respect to its psychometric properties but also to its utility in basic research and in criminal justice. However, an important issue is the extent to which the strong reliability of PCL-R assessments conducted for basic or applied research extends to areas in which the assessments have direct implications for an individual, such as in civil commitment proceedings, parole decisions, treatment options, and so forth.

Recently, there has been concern that the reliability of PCL-R scores is not as high in "the field" as it is in a research context (Boccaccini, Chevalier, Murrie, & Varela, 2017; DeMatteo et al., 2014). It is important to specify what the "field" means, in order not to conflate routine institutional assessments with adversarial bias and allegiance effects. Each has real-life implications for the offenders and patients, and for society, and assessments must meet strict professional and ethical standards.

Institutional Settings

Ismail and Looman (2018) compared the PCL-R scores of an unselected sample of 175 male offenders independently assessed by experienced clinicians two to three years apart, once at a Federal Assessment Center (T1) and later at a Federal Regional Treatment Center (T2). The mean PCL-R score remained about the same from T1 to T2. Rater reliability (ICC_1) generally was higher than that reported in the PCL-R manual (Hare, 2003, Table 5.1): 0.90 for Total score, 0.78 for Factor 1, 0.90 for Factor 2, and between 0.76 and 0.93 for the four first-order facets (first-order latent variables).

Neumann, Hare, Pardini, and Brand (2016) analyzed a data set of 577 forensic psychiatric patients assessed on two occasions (average time between assessments = 16.5 months) by different well-trained and experienced clinicians from the Expertise Centre for Forensic Psychiatry (EFP) in the Netherlands. The mean PCL-R Total and Factor scores for this patient sample were virtually identical across the T1 and T2 assessments. ICC1 values between T1 and T2 were uniformly high: 0.86 for Total score, and 0.84, 0.74, 0.86, and 0.90 for the Interpersonal, Affective, Lifestyle, and Antisocial facets, respectively.

Dåderman and Hellström (2018) reported similar results in a study of Swedish forensic psychiatric patients. The ICC (single rater, absolute agreement) was 0.89 for the Total score, 0.82 for Factor 1, 0.88 for Factor 2, and 0.78 to 0.86 for the four facets. They stated: "These results stand in contrast to lower reliabilities found in a majority of field studies" (p. 234).

Adversarial Settings

In sentencing and preventative detention/civil commitment hearings, experts hired by the prosecution and the defense often provide widely different PCL-R assessments. This adversarial, allegiance, or *hired gun* effect, of course, is not specific to assessments of psychopathy, but the widespread use of the PCL-R for dangerous offender (DO; Canada), sexually violent predator (SVP), and capital sentencing cases in the United States has drawn considerable attention from the legal system (Blais, 2015; Boccaccini et al., 2017; DeMatteo et al., 2014). In their survey of U.S. case law, DeMatteo et al. (2014) identified 348 cases involving the PCL-R from 2005 to 2011. They noted that among "life with parole" inmates in California, "PCL-R scores are a powerful predictor of release-decision-making" (p. 102). They further noted that the PCL-R "appears to be the most widely used measure of psychopathic traits in forensic settings around the world," that it is primarily a "prosecution tool," that challenges to its admissibility "were rare and typically unsuccessful," and that "on average, prosecution examiners reported PCL-R scores that were 7 points higher than defense examiners" (p. 96). Other investigators have reported similar discrepancies between the PCL-R scores of prosecution and defense experts (e.g., Blais & Forth, 2014). This is hardly surprising, given the adversarial/allegiance nature of prosecution/defense testimony, and the opportunity for either side to engage in tactics that will give them an advantage. At the same time, experts differ in the professional standards they hold, their competencies to conduct proper evaluations, and in the amount and nature of the information they have concerning a client.

In their study of SVP evaluators in Texas, Boccaccini, Murrie, Rufino, and Gardiner (2014, p. 343) noted:

> One important difference between field studies and non-field studies is that researchers typically require evaluators to complete intensive PCL-R training—and even complete formal reliability checks—before scoring for a non-field study, whereas there is no such requirement for routine practice in the field.

This certainly is true with respect to many clinicians in private practice who do evaluations for clients or the courts. However, many institutions and their clinicians do take great pains to conduct psychological assessments carefully and competently, with periodic reliability checks.

Boccaccini et al. (2014) obtained "... indirect but positive support for the benefits of PCL-R training" (p. 337) among SVP evaluators, but noted that other factors are involved in determining their ability to provide reliable PCL-R scores. It is clear that formal PCL-R training by itself is not enough and that subsequent "hands on" experience, consultation, and oversight are required (see Note 2). For example, Blais, Forth, & Hare (2017) examined the PCL-R scores of a large number of clinicians and researchers who had taken a Darkstone PCL-R workshop and who subsequently scored three videotaped case histories and extensive case-history information. The inter-rater reliability (ICC) was very high for Total and Factor 2 scores but barely acceptable for Factor 1 scores. The Total scores of about 20% of the raters were very divergent (more than one standard error of measurement; see the next section) from the criterion scores established by Darkstone. It is likely that personal

interactions (interviews) between raters and offenders and patients facilitate the reliable scoring of interpersonal and affective items in Factor 1. That is, the use of videotaped interviews to score Factor 1 items appears to be a relatively poor proxy for what a skilled clinician does in face-to-face interviews.

The adversarial system is firmly entrenched in many jurisdictions, including Canada and the United States, and it is unlikely to change anytime soon. In many cases, the problem is related to limited forensic training and experience, improper use of the PCL-R, failure to keep abreast of the research literature, and the inability or unwillingness of some forensic experts to adhere to established professional and ethical standards of practice (Boccaccini et al., 2017; Hare et al., 2018; Lyon, Ogloff, & Shepherd, 2016). However, there are several ways to ameliorate the problem (Dror & Murrie, 2018; Hare, 2016, pp. 29–30).

For example, Hare et al. (2018, p. 58) suggested:

> Solutions to this problem are difficult to arrive at but must include providing better information to judges and lawyers, requiring higher professional and ethical training and standards of experts, and ensuring that testimony by experts is considered in an informed manner by court representatives (see Dror & Murrie, 2018). There is no justifiable reason why experienced clinicians cannot provide PCL-R scores that are as reliable as those obtained by researchers and their students (see Note 1).

The use of expert panels or court-appointed experts "may be increasingly useful to help keep expert 'hired guns' honest or, at the least, more restrained in their opinions" (McCahey & Proman, 2011).[4] Judges, lawyers, and experts should become familiar with the strengths and limitations, uses and misuses, of the PCL-R and other clinical/forensic tools. Experts should be required to justify their ratings of individual items. Many jurisdictions have a code of conduct for expert witnesses, which experts should follow. For example, the Federal Court Rules (SOR/98–106) for Canada has Rule 52.2:

> 1. An expert witness named to provide a report for use as evidence, or to testify in a proceeding, has an overriding duty to assist the Court impartially on matters relevant to his or her area of expertise. 2. This duty overrides any duty to a party to the proceeding, including the person retaining the expert witness. An expert is to be independent and objective. An expert is not an advocate for a party. (Minister of Justice, Government of Canada, 2015, p. 421)

Not all experts are equal in what they do. Some may not "have the personality, astuteness, impartiality, judgment, and interpersonal skills needed to collect, integrate, and interpret the extensive information involved in complex psychological evaluations" (Hare, 2016, p. 30). A valuable guide for forensic assessments is the Hierarchy of Expert Performance (HEP; Dror & Murrie, 2018).

There is a substantial amount of empirical evidence that psychopathy, as measured by the PCL-R and its derivatives, is a predictor of recidivism and violence in prison, forensic psychiatric, and civil psychiatric populations. Indeed, the PCL-R is one of the most generalizable of the risk factors identified thus far, and for this reason, it is included in a variety of actuarial and structured-clinical risk assessment procedures. Although psychopathy is not the only risk factor for recidivism and violence, it is unusually pervasive and too

important to ignore, particularly with respect to violence. Treatment and management are difficult, time-consuming, and expensive, but new initiatives based on current theory and research on psychopathy and the most effective correctional philosophies may help to reduce the harm done by psychopaths.

Criminal Justice and the PCL-R

Although developed to measure a clinical construct, the PCL-R has become one of the most widely used psychological instruments in forensic psychiatry and practice. In many cases, the PCL-R and its derivatives are at least as predictive of general and violent offending in a variety of contexts as are purpose-built risk tools (Blais & Bonta, 2015; de Vogel & Lancel, 2016; Hemphill & Hare, 2004; Hilton, Harris, Rice, Houghton, & Eke, 2008; Krstic et al., 2017; Leistico, Salekin, DeCoster, & Rogers, 2008; Mokros, Vohs, & Habermeyer, 2014; Olver & Wong, 2015; Rice & Harris, 2013; Robertson & Knight, 2014; van der Put, Gubbels, & Assink, 2019; Vitacco, Gonsalves, Tomony, Smith, & Lishner, 2012; Yang, Wong, & Coid, 2010). The combination of a high PCL-R score and evidence of deviant sexuality is a strong predictor of sexual reoffending (Harris et al., 2003; Hawes, Boccaccini, & Murrie, 2013).

Here I address a common finding in the risk literature that Factor 2 (Lifestyle/Antisocial), especially its Antisocial facet, is more predictive of outcomes of interest to the criminal justice system than is Factor 1 (Interpersonal/Affective). For this reason, some commentators see no role for Factor 1 in criminal justice, a serious mistake in my view. It is important to remember that I did not design the PCL-R as a risk tool, and that the PCL-R defines psychopathy by elevated scores on both Factor 1 and Factor 2. Its role is as part of a complete clinical/behavioral work-up to assist decision-makers in a variety of contexts in which psychopathic traits and behaviors have clinical or probative relevance (Gacono, 2016; Hare, 2003).

The relative predictive utility of the factors and facets depends on what they predict and on the appropriateness and sophistication of the methods used to analyze the data. For example, Factor 1 is at least as predictive as Factor 2 of in-patient violence (Langton, Hogue, Daffern, Mannion, & Howells, 2011; Vitacco et al., 2012), sexual violence (Harris, Rice, Hilton, Lalumière, & Quinsey, 2007), time to drop out from therapy (Brunner at al., 2019; Olver & Wong, 2011), instrumental violence (Blais, Solodukhin, & Forth, 2014), and time to first violence (Wilson, 2003). Olver, Lewis, and Wong (2013) reported that, during a five-year post-release period for 150 violent offenders, Factor 1 (especially the Affective facet) was *more* predictive of violence than was Factor 2. Similarly, Olver and Wong (2011) reported that the strongest predictors of dropout from treatment among these offenders were the Affective facet of Factor 1 and never being married. Factor 1 also appears to be important in explaining and predicting domestic violence (Swogger, Walsh, & Kosson, 2007).

Many conclusions about the relative predictive utility of the PCL factors and facets derive from zero-order or partial correlations with some specific outcome, such as recidivism, or from regression analyses. Alternative latent-variable (e.g., SEM) analyses provide evidence that the interpersonal and affective facets may contribute significantly to prediction of clinical criteria, including in-patient violence, treatment outcome, aggression, criminal

recidivism, and other externalizing psychopathology (Hill et al., 2004; Krstic et al., 2017; Olver et al., 2013; Vitacco & Kosson, 2010; Vitacco et al., 2005). In their latent-variable analyses, Vitacco et al. (2005) performed a CFA on the 1136 civil psychiatric patients in the MacArthur Violence Risk Assessment Study, and obtained good fit for a four-factor model (Table 4.1, right panel). They then used SEM to understand the relationships among the four facets and violence at 20-week follow-up. The SEM results indicated that the latent-variable correlation with violence was 0.41 for the Affective facet and 0.40 for the Antisocial facet. Several recent articles and chapters (Hare et al., 2013; Mokros et al., Chapter 5 this volume; Neumann, Vitacco et al., 2016) show how the four dimensions of psychopathy make unique contributions to the prediction of various variables of interest to the criminal justice system, including time to first violent offense, type of sexual offending, early dropout from therapy, and instrumental violence. For example, using PCL-R data for 958 sex offenders described by Robertson and Knight (2014), Krstic et al. (2017) reported that the latent-variable correlation with violence was 0.33 for the Affective factor and 0.35 for the Antisocial factor.

The relevance and importance of the PCL factors and facets in criminal justice depend on the context and way in which they are used. As indicated by Neumann, Vitacco, et al. (2016) and by Mokros et al. (Chapter 5 this volume), the *pattern* of facet scores may be particularly informative to criminal justice. Parenthetically, all four PCL facets play important roles in the neuroscience of psychopathy (e.g., Crooks et al., 2018; Glenn & Raine, 2014; Lushing et al., 2016; Chapter 11 this volume).

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Notes

1 A model for the training of researchers is provided by Brown, Dargis, Mattern, Tsonis, and Newman (2015, p. 1036). Trained undergraduate students, graduate students, and professional staff completed the PCL-R interviews and ratings. All interviewers were required to complete an intensive training process that included a thorough education of the following: the construct of psychopathy, the development of the PCL-R and PCL-R manual, the Factor structure of the PCL-R, and principles of PCL-R ratings. After this education, new interviewers shadowed experienced interviewers for several months until a very high level of reliability was reached. Dual ratings for PCL-R total scores were available for 559 participants. Using interclass correlation, the inter-rater reliability for these ratings was 0.915.

2 In three meta-analyses, Derefinko (2015) concluded that the findings suggest that although psychopathic individuals have deficits in inhibition/constraint, they do not necessarily exhibit a consistent absence of negative affect ...[and] that while constraint composes a large part of psychopathy assessments, it is less clear how much anxiety lends to the construct (p. 693).

In a meta-analysis Hoppenbrouwers, Bulten, & Brazil (2016) concluded that fearlessness in psychopathy may involve more a failure to respond to threat cues than a subjective feeling of fear.

3 Portions of this section are adapted from Hare (2016, pp. 29–30).

4 McCahey and Proman (2011) wrote that the United States Federal Rule of Evidence 706 "provides a means to slice through the fog of conflicting expert testimony and obtain unbiased testimony from a court-appointed expert. Although it has been infrequently invoked in the nearly 40 years since its enactment, Rule 706 may be utilized more often in the future as the issues that judges and juries are called upon to decide become increasingly more complex and the testimony of litigants' experts become more partisan. Further, the use Rule 706 court-appointed experts, or the threat thereof, may be increasingly useful to help keep expert 'hired guns' honest or, at the least, more restrained in their opinions. Rule 706 experts increasingly may be the tool needed to help ensure that the adversary system allows the truth to emerge."

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